

FIELDCOMMS

Incident Management System v1.0

BETA TEST CHECKLIST

ESV Field Testing & Verification

RACES · ARES · Starcom · K9ESV · MCESV / MCEMA

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1 · System & Infrastructure

Access, services, Wi-Fi AP, and health monitoring

1.1 Access & Network

P	F	FEATURE / FUNCTION	NOTES
		Dashboard loads at http://192.168.50.1 on a device connected to EMCOMM-NET	Try phone, tablet, laptop
		SSID "EMCOMM-NET" visible and password accepted	WPA2
		Dashboard loads at http://fieldcomms.local (mDNS alias)	
		Dashboard loads at http://emcomm.local (mDNS alias)	
		Single router coverage verified — EMCOMM-NET reachable throughout the deployment area	
		All dashboard cards load without 404 errors	Click each card
		UTC clock in nav bar updates every second	

1.1b AiMesh Coverage Extension (standard 2-node deployment)

P	F	FEATURE / FUNCTION	NOTES
		Both RT-BE58 Go mesh nodes powered on and showing Connected in primary router AiMesh menu	http://192.168.50.254 → AiMesh → AiMesh Node — both must show Connected
		Mesh Node 1 backhaul type shows Ethernet (UniFi Switch Port 11)	Wired backhaul confirms CAT 6 cable is seated at switch and node LAN port
		Mesh Node 2 backhaul type shows Ethernet (UniFi Switch Port 12)	Wired backhaul confirms CAT 6 cable is seated at switch and node LAN port
		EMCOMM-NET and http://192.168.50.1 reachable from Node 1 coverage zone	Walk to secondary room/wing with a phone
		EMCOMM-NET and http://192.168.50.1 reachable from Node 2 coverage zone	Walk to third coverage zone with a phone
		Device roams between nodes without reconnecting when moving through venue	No Wi-Fi prompt, no password re-entry — seamless roam
		Pi remains at 192.168.50.1 throughout — not affected by which node is connected	

1.1c WAN Connectivity (InstyConnect + Starlink failover)

P	F	FEATURE / FUNCTION	NOTES
		InstyConnect Drum powered on and connected to ASUS WAN port via PoE cable	Check Drum LED status — green or white = connected to carrier
		ASUS WAN port shows Connected status with InstyConnect IP address	http://192.168.50.254 → WAN → Internet Status — must show Connected
		WAN Status dashboard shows active source as InstyConnect Cellular	http://192.168.50.1/wan-status.html — WAN card must be green
		Carrier shown on WAN status page (T-Mobile or Verizon)	Confirms dual-carrier multi-network plan is active
		NWS weather alerts loading on dashboard (requires WAN)	Alerts section populates — confirms cellular internet is reaching external APIs
		Starlink dish powered on and connected to ASUS USB WAN port via adapter	Dish status LED — may take up to 3 minutes to acquire signal after power-on
		ASUS Dual WAN configured: Primary = WAN port, Secondary = USB, Mode = Failover	http://192.168.50.254 → WAN → Dual WAN — verify all three settings
		Failover test: unplug InstyConnect PoE cable — ASUS switches to Starlink within 60 seconds	Watch ASUS WAN status page. WAN card on dashboard should turn blue.
		Failback test: reconnect InstyConnect PoE cable — ASUS switches back to cellular within 60 seconds	WAN card on dashboard returns to green. InstyConnect is primary again.
		Reconnect InstyConnect PoE cable after failover test	Restore normal WAN configuration

1.1d AMPRNet / 44Net Gateway

P	F	FEATURE / FUNCTION	NOTES
		Gateway Pi powered on at 192.168.50.2 (check green power LED)	Gateway Pi must be on the same UniFi switch as FieldComms Pi
		AMPRNet Gateway card on FieldComms dashboard shows green (tunnel UP)	http://192.168.50.1 — bottom connectivity section — amprgate card must be green
		Gateway status page loads at http://192.168.50.2:9000	Should show callsign login page
		Callsign login works — enter a valid FCC callsign at http://192.168.50.2:9000	Dashboard must load showing tunnel status after login
		Tunnel status shows UP with a 44.x.x.x/29 AMPRNet address assigned	Confirms WireGuard tunnel to amprgw.ampr.org is established
		Last handshake time shown — within the past 5 minutes	Confirms tunnel is actively maintained
		Access log entry created — check <code>/var/log/amprgate-access.log</code> on gateway Pi	SSH to gateway Pi: <code>tail /var/log/amprgate-access.log</code> — must show LOGIN-OK with callsign
		Tunnel control blocked from network — port 9001 not reachable from EMCOMM-NET	From any EMCOMM-NET device: <code>curl http://192.168.50.2:9001</code> must return connection refused
		WAN Status dashboard shows AMPRNet tunnel state	http://192.168.50.1/wan-status.html — AMPRNet section shows tunnel UP

1.2 Pre-Flight Health Check (<http://192.168.50.1/preflight.html>)

P	F	FEATURE / FUNCTION	NOTES
		Pre-flight checklist page loads	
		All checklist items visible and checkable	
		Pass/Fail verdict renders correctly (GO / CAUTION / NO-GO)	
		Export Report generates a printable summary	

1.3 Dashboard Health Monitor (port 5051)

P	F	FEATURE / FUNCTION	NOTES
		Dashboard health panel shows CPU temperature	
		Dashboard health panel shows memory usage	
		Dashboard health panel shows disk usage	
		Service status shown for: nginx, FCC Lookup API, Health Monitor	
		Service status shown for: Graywolf APRS, Pat Winlink, Kiwix	
		Service status shown for: GPS Daemon, NTP/Chrony, Dead Man's Switch, ICS Platform	
		Internet connectivity indicator shown	
		GPS status shown (fix / no fix)	Requires USB GPS
		Health data refreshes automatically	

1.4 Dashboard Mode Switcher

P	F	FEATURE / FUNCTION	NOTES
		Three mode buttons visible in the mode bar: ■ Amateur Radio · ■ Starcom / Public Safety · ■ ICS	
		Mode selection persists across browser sessions (saved to localStorage)	
		Amateur mode — Net Ops: Net Control Logger, Observer Mode, JS8Call, APRS Tactical Map	
		Amateur mode — Reference: Callsign Lookup, HF Propagation, Repeaters, Grid Square, Cheat Sheets	
		Amateur mode — Forms: NTS Radiogram, ICS-213, ICS-214, Print Center, Reference Library	
		Amateur mode — Safety: Dead Man's Switch, Roster, Pre-Flight Checklist	
		Starcom mode — Net Ops: Starcom Net Logger, Weather Net, SAR Net, Observer Mode	Starcom is now its own dedicated mode
		Starcom mode — Resources: Resource Tracking Map, Resource Board, Member Roster, Facilities	
		Starcom mode — ICS: ICS Platform, ICS-213, ICS-214, Print Center	
		ICS mode — Platform: ICS Command, Operations, Planning, Logistics, Finance, Planning P	
		ICS mode — Resources: Tactical Map, Resource Map, Resource Board, Roster, Facilities	
		ICS mode — Forms: ICS-213, ICS-214, ICS-309, Print Center, Reference Library	
		ICS mode — Comms: Net Control, Starcom, Winlink Form Import, Repeater Database, Cheat Sheets	
		URL param ?mode=starcom loads dashboard in Starcom mode directly	

2 · Net Control Logging

Amateur net control and Starcom net logging

2.1 Amateur Net Control (<http://192.168.50.1/netcontrol.html>)

P	F	FEATURE / FUNCTION	NOTES
		Create a new net — name, type, frequency, drill flag	
		Net appears in the net list and can be selected	
		Log a net entry (callsign, name, status, location, remarks)	
		Entry appears in the log with UTC timestamp	
		Log Traffic button adds a traffic / message entry	
		Check-In from roster chip works (pre-populated callsign)	
		Remove an entry from the log	
		Promote an operator to the roster from a net entry	
		Export ICS-309 opens a printable comms log for the selected net	
		Export JSON downloads the net data as a file	
		Close net marks it as closed and archives it	
		Multiple nets open simultaneously — each on its own tab	
		Observer link copies the read-only URL to clipboard	

2.2 Starcom Net Logger (<http://192.168.50.1/starcom.html>)

P	F	FEATURE / FUNCTION	NOTES
		Create a new Starcom net (name, talkgroup, channel, dispatch center)	
		Log a check-in by Radio ID and Unit Name	
		Unit status cycles correctly: Available → Deployed → Staging → Out of Service	
		Multiple simultaneous nets — each net badge shown in active nets panel	Core Starcom feature
		Clicking a net badge switches to that net log	
		Weather Net quick-launch card opens pre-configured weather net	
		SAR Net quick-launch card opens pre-configured SAR net	
		Export ICS-309 works for the Starcom net	
		Close net archives all entries	

3 · Resource & Personnel Management

3.1 Member Roster (<http://192.168.50.1/roster.html>)

P	F	FEATURE / FUNCTION	NOTES
☐	☐	Member list loads from server database	
☐	☐	Add a new member (name, callsign, role, certifications)	
☐	☐	Edit an existing member	
☐	☐	Delete a member	
☐	☐	Check in a member to the active activation	
☐	☐	Walk-in check-in (non-roster member)	
☐	☐	Export CSV downloads the full roster	
☐	☐	Import CSV loads a roster from file	
☐	☐	Export activation log downloads the check-in record	
☐	☐	Filter / search the roster	

3.2 Resource Board (<http://192.168.50.1/resources.html>)

P	F	FEATURE / FUNCTION	NOTES
☐	☐	Resource cards load from server	
☐	☐	Add a resource (name, type, owner, location, status)	
☐	☐	Status badge cycles through all 5 states: Available / Assigned / Staging / OOS / Demobilized	
☐	☐	Edit a resource	
☐	☐	Delete a resource	
☐	☐	Filter by type and status	
☐	☐	Stats strip at top updates live	

3.3 Facilities (<http://192.168.50.1/facilities.html>)

P	F	FEATURE / FUNCTION	NOTES
☐	☐	Seeded facilities load (McHenry County EOC, etc.)	
☐	☐	Add a new facility (name, type, address, lat/lon)	
☐	☐	Edit a facility	
☐	☐	Delete a facility	
☐	☐	Copy address to clipboard	
☐	☐	Export CSV	
☐	☐	Facility detail modal shows all fields	

4 · Maps & Tactical Display

4.1 APRS Tactical Map (<http://192.168.50.1/tactical.html>)

P	F	FEATURE / FUNCTION	NOTES
☐	☐	Map loads with offline tile layer (USGS Imagery+Topo)	
☐	☐	APRS stations appear as markers from Graywolf	Requires TNC/radio
☐	☐	Clicking a station marker shows callsign and details	
☐	☐	Resource overlays from Resource Board appear on the map	
☐	☐	Draw marker, polygon, circle, and route overlays	
☐	☐	KML export downloads current overlays	
☐	☐	Range ring draws around current position	
☐	☐	Station list sidebar filters by callsign	
☐	☐	Station age threshold hides old stations	
☐	☐	Tile set selector switches between USGS/OSM/Esri	

4.2 Resource Map / Starcom Map (<http://192.168.50.1/resmap.html>)

P	F	FEATURE / FUNCTION	NOTES
☐	☐	Map loads with offline tiles	
☐	☐	Add a unit to the map (place a marker)	
☐	☐	Edit a unit's details	
☐	☐	Remove a unit from the map	
☐	☐	Drag unit marker to new position — coordinates update	
☐	☐	Draw a zone polygon	
☐	☐	Zone listed in Zones tab	
☐	☐	Clear zones removes all drawn zones	
☐	☐	Unit status colors display correctly	

4.3 Grid Square Calculator (<http://192.168.50.1/grid.html>)

P	F	FEATURE / FUNCTION	NOTES
☐	☐	Convert a lat/lon to Maidenhead grid square	
☐	☐	Convert a grid square to lat/lon	
☐	☐	Calculate distance and bearing between two grid squares	
☐	☐	Use My Location gets device GPS coordinates	Mobile device

5 · ICS Platform

<http://192.168.50.1/ics/>

5.1 Incident Management

P	F	FEATURE / FUNCTION	NOTES
☐	☐	Create a new incident (name, type, location, IC, op period)	
☐	☐	Incident becomes the active incident for all ICS pages	
☐	☐	Advance operational period increments the period number	
☐	☐	Close / archive an incident	
☐	☐	ICS navigation tabs visible: Command / Operations / Planning / Logistics / Finance / Planning P	

5.2 Command Section (ics/command.html)

P	F	FEATURE / FUNCTION	NOTES
☐	☐	100-item objectives dropdown present and organized into 13 groups	
☐	☐	Selecting an objective pre-fills the text field	
☐	☐	Editing and adding a custom objective works	
☐	☐	Objectives list correctly and can be deleted	
☐	☐	Safety message field accepts input	
☐	☐	Situation report fields save correctly	
☐	☐	Save persists data for the active incident	
☐	☐	MCESV/MCEMA-specific objectives appear in the last group	

5.3 Operations Section (ics/operations.html)

P	F	FEATURE / FUNCTION	NOTES
☐	☐	T-card board loads for the active incident	
☐	☐	Add a resource assignment (callsign, assignment, status)	
☐	☐	Drag-and-drop T-cards between status columns	
☐	☐	Set resource status: Available / Assigned / Staging / Out of Service / Released	
☐	☐	Delete a resource	
☐	☐	Open resource detail panel and update assignment	

5.4 Planning Section (ics/planning.html)

P	F	FEATURE / FUNCTION	NOTES
☐	☐	IAP tracker shows all ICS forms with completion checkboxes	
☐	☐	Resource status table shows counts by type	
☐	☐	Documents tab lists linked incident documents	
☐	☐	Save persists planning section data	

5.5 Logistics Section (ics/logistics.html)

P	F	FEATURE / FUNCTION	NOTES
☐	☐	ICS-205 comms plan pre-filled with McHenry County channels	
☐	☐	Supply tracking entries can be added and deleted	
☐	☐	Meal log entries can be added	
☐	☐	Check-in list (ICS-211) accepts entries	

5.6 Finance / Admin Section (ics/finance.html)

P	F	FEATURE / FUNCTION	NOTES
☐	☐	Cost tracking entries can be added and deleted	
☐	☐	Time tracking entries can be added and deleted	
☐	☐	Procurement entries can be added and deleted	
☐	☐	Export CSV downloads the finance data	

6 · ICS Forms

6.1 ICS-213 General Message (<http://192.168.50.1/ics213.html>)

P	F	FEATURE / FUNCTION	NOTES
☐	☐	Form fields all accept input (To, From, positions, date, time, subject, message, reply)	
☐	☐	Auto-fill date/time populates current UTC	
☐	☐	Save to Log saves the form to the local log	
☐	☐	Form log shows saved forms, most recent first	
☐	☐	Load a saved form from the log	
☐	☐	Delete a saved form	
☐	☐	Print button opens browser print dialog	
☐	☐	Clear Form resets all fields	

6.2 ICS-214 Activity Log (<http://192.168.50.1/ics214.html>)

P	F	FEATURE / FUNCTION	NOTES
☐	☐	Header fields accept input (incident, unit, leader, op period)	
☐	☐	Add Personnel adds a row (name, position, agency)	
☐	☐	Remove Personnel removes a row	
☐	☐	Add Entry adds an activity log row	
☐	☐	Auto Timestamp populates current UTC on a row	
☐	☐	Remove Entry removes an activity row	
☐	☐	Generate Form renders the print-ready ICS-214	
☐	☐	Print works correctly	
☐	☐	Save Local persists the form on the device	
☐	☐	Clear Form resets everything	

6.3 ICS-309 Communications Log (<http://192.168.50.1/ics309.html>)

P	F	FEATURE / FUNCTION	NOTES
☐	☐	Header fields accept input (incident name, op period, radio operator, station/net)	
☐	☐	Add (timestamp now) inserts a row with current UTC time	
☐	☐	Log rows accept From, To, and Subject entries	
☐	☐	Delete a log row	
☐	☐	Generate Form renders the print-ready ICS-309	
☐	☐	Save to Incident files the log with the active incident	

6.4 Planning P (<http://192.168.50.1/ics/planningp.html>)

P	F	FEATURE / FUNCTION	NOTES
		Page loads with Planning P reference diagram image on the left	
		All 15 phase buttons present in the center panel	
		Phase buttons organized in 5 color-coded groups (gray/yellow/red/green/teal)	
		Clicking a phase highlights it and populates the right detail panel	
		Detail panel shows Standard Agenda, Required Forms, and Who Should Attend	
		Form chips in detail panel open the correct form page when clicked	
		Generate briefing sheet opens printable one-page cover for the selected phase	
		Briefing sheet includes phase name, agenda, forms table, and attendance table	

6.5 NTS Radiogram (<http://192.168.50.1/nts.html>)

P	F	FEATURE / FUNCTION	NOTES
		Auto-fill date/time populates current date and time	
		Auto-fill number generates a sequence number	
		All preamble fields accept input (number, precedence, handling, origin, check, place, time)	
		Address and text fields accept input	
		Generate Radiogram renders the formatted radiogram	
		Save to Log saves the radiogram	
		Load from Log restores a saved radiogram	
		Delete a saved radiogram	
		Copy Text copies the radiogram to clipboard	
		Clear Form resets all fields	

7 · Winlink

7.1 Pat Winlink Backup (<http://192.168.50.1:8090>)

P	F	FEATURE / FUNCTION	NOTES
		Pat Winlink loads at port 8090	
		Pat dashboard card on FieldComms opens the correct URL	
		Can compose and send a message (requires TNC/radio or internet)	Optional
		Pat service shows as running in Health Monitor	

7.2 Winlink Form Import (<http://192.168.50.1/winlink-import.html>)

P	F	FEATURE / FUNCTION	NOTES
		Page loads from dashboard ICS mode card	
		Drop zone accepts an RMS_Express_Form XML file	
		Paste text area accepts pasted XML or message text	
		Parse Form Data button processes the input	
		ICS-213 form is detected and labeled correctly	
		ICS-214 form is detected and labeled correctly	
		ICS-309 form is detected and labeled correctly	
		Unknown form type shows a warning (not an error)	
		All mapped fields appear in the editable review grid	
		Incident dropdown loads active incidents	
		Archive to Incident saves the form to the incident record	
		Open in ICS form routes ICS-213/214/309 to the correct form page	

8 · Callsign & Radio Tools

8.1 FCC Callsign Lookup (<http://192.168.50.1/callsign.html>)

P	F	FEATURE / FUNCTION	NOTES
☒	☒	Quick lookup by callsign returns name, license class, address, grid, FRN	
☒	☒	Lookup works offline (from local FCC database)	
☒	☒	Advanced search by name, city, state, license class works	
☒	☒	Add to Roster button adds the licensee to the member roster	
☒	☒	FCC database record count and last-update date shown at bottom	

8.2 Repeater Database (<http://192.168.50.1/repeaters.html>)

P	F	FEATURE / FUNCTION	NOTES
☒	☒	Page loads and shows the three source tabs (Offline File, Server API, Sample)	
☒	☒	Sample data tab shows SAMPLE-1/2/3 placeholders with "not a real repeater" warning	
☒	☒	Offline File tab: drag-and-drop a RepeaterBook CSV loads and displays repeaters	
☒	☒	Filter by band (2m, 70cm, etc.) works	
☒	☒	Filter by state works	
☒	☒	Filter by ARES / SKYWARN / RACES works	
☒	☒	Filter by mode (FM, DMR, C4FM, etc.) works	
☒	☒	Sort by callsign, frequency, and city works	
☒	☒	Clicking a repeater row opens the detail panel	

8.3 HF Propagation (<http://192.168.50.1/propagation.html>)

P	F	FEATURE / FUNCTION	NOTES
☒	☒	Page loads and attempts to fetch solar data from HamQSL	Requires internet
☒	☒	SFI, K-index, A-index displayed	
☒	☒	Band condition indicators show (10m–80m)	
☒	☒	Offline fallback message shown when no internet	

9 · Reference, Print & Kiwix

9.1 Reference Library (<http://192.168.50.1/refs.html>)

P	F	FEATURE / FUNCTION	NOTES
☒	☒	Document list loads from server	
☒	☒	Upload a document (PDF, DOCX, image) via drag-and-drop	
☒	☒	Upload via Browse button works	
☒	☒	PDF thumbnail generated after upload	
☒	☒	Filter by category works	
☒	☒	Search by keyword finds matching documents	
☒	☒	Edit document metadata	
☒	☒	Download a document	
☒	☒	Delete a document	

9.2 Cheat Sheets (<http://192.168.50.1/cheatsheets.html>)

P	F	FEATURE / FUNCTION	NOTES
☒	☒	Phonetic alphabet table displays correctly	
☒	☒	Q-codes reference table displays	
☒	☒	Band plan reference displays	
☒	☒	CTCSS/DCS tones table displays	
☒	☒	ICS position titles reference displays	
☒	☒	Print button sends page to printer	

9.3 Print Center (<http://192.168.50.1/printcenter.html>)

P	F	FEATURE / FUNCTION	NOTES
☐	☐	ICS-213 preview and print works	
☐	☐	ICS-214 preview and print works	
☐	☐	NTS Radiogram preview and print works	
☐	☐	Pre-Flight Checklist preview and print works	
☐	☐	Phonetic Alphabet pocket card prints	
☐	☐	Q-Codes & Prowords reference prints	
☐	☐	Net Control Log ICS-309 export and print works	
☐	☐	Starcom Net Log ICS-309 export and print works	
☐	☐	Roster / Member Directory prints	
☐	☐	Resource Board status print works	
☐	☐	Incident cover sheet (Generate Cover) creates a cover page	
☐	☐	Operator Access Cards PDF generates (Avery 5371 format)	
☐	☐	Print button on ICS-213 opens browser print dialog	
☐	☐	If CUPS installed: http://192.168.50.1:631 loads CUPS admin UI	Option B — USB printer
☐	☐	If CUPS installed: USB printer visible in CUPS printer list	Option B
☐	☐	If CUPS installed: test page prints successfully from CUPS UI	Option B
☐	☐	If CUPS installed: Windows device can discover and print to shared printer	Option B
☐	☐	If network printer: printer visible in browser print dialog from EMCOMM-NET device	Option C

9.4 Kiwix Offline Library (<http://192.168.50.1:8081>)

P	F	FEATURE / FUNCTION	NOTES
☐	☐	Kiwix loads at port 8081	
☐	☐	At least one ZIM file (WikiMed, Wikipedia, iFixit, or Wikivoyage) is present	
☐	☐	Can search and retrieve articles	
☐	☐	Kiwix dashboard card opens the correct URL	

10 · Dead Man's Switch<http://192.168.50.1/deadmans.html>**10.1 Dead Man's Switch**

P	F	FEATURE / FUNCTION	NOTES
☐	☐	Page loads and shows active nets	
☐	☐	Arm button activates monitoring for a net	
☐	☐	Countdown timer displays and counts down	
☐	☐	Reset button resets the countdown without disarming	
☐	☐	Disarm button stops monitoring	
☐	☐	DMS service shown as active in health monitor	

11 · Operator Identity

11.1 Identity System

P	F	FEATURE / FUNCTION	NOTES
☐	☐	First load prompts for operator identity (callsign/Radio ID/name, role)	
☐	☐	Identity persists across page reloads and navigation	
☐	☐	Callsign/ID shown in dashboard header after setup	
☐	☐	Role shown correctly: NCS / Operator / Liaison / EC	
☐	☐	Identity can be changed / reset	
☐	☐	Observer page does NOT prompt for identity (read-only)	

12 · System Services

Verify via `systemctl` status on the Pi, or via the dashboard health monitor.

12.1 Service Status

P	F	FEATURE / FUNCTION	NOTES
		fcc-lookup.service — main API running on port 5050	
		health-monitor.service — health API running on port 5051	
		wan-monitor.service — WAN status monitor polling InstyConnect and Starlink	
		amprgate-poll.service — polls gateway Pi every 30s for 44Net status	
		ics-platform.service — ICS API running on port 5055	
		fieldcomms-refs.service — reference library API running on port 5056	
		fieldcomms-tiles.service — tile server running on port 8083	
		deadmans.service — Dead Man's Switch service running	
		nginx.service — web server running on port 80	
		kiwix.service — Kiwix library running on port 8081	
		pat.service — Pat Winlink running on port 8090	
		yaac.service — YAAC APRS client running on port 8082	If APRS installed
		graywolf.service — Graywolf APRS running on port 8080	If APRS installed
		cups.service — CUPS print server running on port 631	If printer installed
		avahi-daemon.service — mDNS/Bonjour running	
		gpsd.service — GPS daemon running (if USB GPS connected)	If GPS installed
		fcc-refresh.timer — weekly FCC database refresh timer active	
		repeater-refresh.timer — monthly repeater database refresh timer active	
		Firewall (ufw) enabled — ports 80, 631, 5050-5056, 8080-8090 open	

12.2 Gateway Pi Services (192.168.50.2 — SSH separately)

P	F	FEATURE / FUNCTION	NOTES
		amprgate-status.service — status page running on port 9000	SSH to 192.168.50.2: systemctl status amprgate-status
		wg-quick@ampr0.service — WireGuard tunnel running	SSH to 192.168.50.2: systemctl status wg-quick@ampr0
		Port 9000 accessible from EMCOMM-NET (callsign login page loads)	From any EMCOMM-NET device: open http://192.168.50.2:9000
		Port 9001 NOT accessible from EMCOMM-NET (connection refused)	From any EMCOMM-NET device: curl http://192.168.50.2:9001 — must fail
		Access log file exists at /var/log/amprgate-access.log	SSH to gateway Pi: ls -la /var/log/amprgate-access.log
		UFW firewall active on gateway Pi — port 9001 blocked externally	SSH to gateway Pi: sudo ufw status — verify 9001 is DENY

Beta Test Sign-Off

Section	Items	Pass	Fail	Not Tested
1 · System & Infrastructure (incl. WAN + 44Net)				
2 · Net Control Logging				
3 · Resource & Personnel				
4 · Maps & Tactical				
5 · ICS Platform				
6 · ICS Forms				
7 · Winlink				
8 · Callsign & Radio Tools				
9 · Reference, Print & Kiwix				
10 · Dead Man's Switch				
11 · Operator Identity				
12 · System Services				
TOTALS				

Failures requiring follow-up:

**Signed off by (name /
callsign):**

Date:

Submit completed checklist (photo or scan) to the project lead. Include this sheet plus any additional failure notes. Reference the user manual for expected behavior of any item.

Operator Workstation Options — Raspberry Pi 500 / 500+ + Raspberry Pi Monitor

FieldComms is a browser-based system — any device with a modern browser and Wi-Fi can connect to EMCOMM-NET and access every tool. The Raspberry Pi 500 paired with the official Raspberry Pi Monitor provides a purpose-built, low-cost, low-power operator workstation that is fully Pi ecosystem compatible and ideal for fixed EOC positions, shelter check-in stations, and net control operator desks. Funding through ARPA-E grants, FEMA BRIC, or 501(c)(3) equipment donations may cover these workstations.

Pi 500 vs Pi 500+ vs Pi 400 — Comparison

SPEC	Pi 400 (2020)	Pi 500 (Dec 2024) ★ Recommended	Pi 500+ (Sept 2025)
Processor	Cortex-A72 @ 1.8 GHz (Pi 4 generation)	Cortex-A76 @ 2.4 GHz (Pi 5 generation)	Cortex-A76 @ 2.4 GHz (Pi 5 generation)
RAM	4 GB DDR4	8 GB LPDDR4X	16 GB LPDDR4X
Storage	microSD only	microSD only (32 GB A2 included)	256 GB NVMe built-in + microSD slot
Keyboard	Membrane	Membrane	Mechanical (Gateron Blue KS-33)
Wi-Fi	802.11ac dual-band	802.11ac dual-band	802.11ac dual-band
Ethernet	Gigabit	Gigabit	Gigabit
USB ports	2× USB 3.0, 1× USB 2.0	2× USB 3.0, 1× USB 2.0	2× USB 3.0, 1× USB 2.0
Display output	2× micro-HDMI	2× micro-HDMI	2× micro-HDMI
Production until	Jan 2028	TBD (current model)	TBD (current model)
Best for FieldComms?	Acceptable — older chip, only 4 GB RAM	✓ YES — Pi 5 chip, 8 GB RAM, current generation	Overkill for browser use Justified only if also running heavy Linux apps locally

Raspberry Pi Monitor — 15.6"

SPEC	DETAIL
Panel	15.6" IPS LCD, 1920×1080 Full HD @ 60 Hz
Video input	Standard HDMI 1.4 (full-size) — requires micro-HDMI to HDMI cable with Pi 500
Audio	2× front-facing stereo speakers, 3.5mm headphone jack
Power	USB-C, 5V/1.5A — can power from Pi 500 USB port at 60% brightness/volume, or from separate USB-C supply for 100% brightness/volume
Mounting	Integrated angle-adjustable kickstand, VESA mount, wall hanger
Dimensions	Slim 15.6" form — similar footprint to a laptop
Colors	Red/White (matching Pi 500 keyboard) or Black
Where to buy	raspberrypi.com · Adafruit · PiShop.us · GigaParts · Amazon

The Pi 500 has micro-HDMI ports; the Raspberry Pi Monitor has a standard full-size HDMI input. You need a micro-HDMI to standard HDMI cable (or adapter). The Pi 500 desktop kit (\$120) includes one. When ordering separately, confirm the cable is included or purchase one — they are available from the same retailers listed above.

Operator Workstation — Bill of Materials (Per Station)

ITEM	MODEL / SPEC	WHERE TO BUY
— Standard Workstation (Pi 500) —		
Raspberry Pi 500 keyboard computer	Pi 500 standalone — Pi 5 chip, 8 GB RAM, 32 GB A2 microSD, keyboard	raspberrypi.com Adafruit · Amazon
Raspberry Pi Monitor	15.6" Full HD IPS, built-in speakers, kickstand, VESA mount	raspberrypi.com GigaParts · Amazon
micro-HDMI to HDMI cable	1m micro-HDMI to standard HDMI (Pi 500 → Monitor) Included in Pi 500 Kit (\$120) — verify before ordering separately	raspberrypi.com Amazon
USB mouse	Any USB or Bluetooth mouse — included in Pi 500 Kit	Amazon
USB-C power supply for Pi 500	Official Raspberry Pi 27W USB-C PD power supply (or equivalent)	raspberrypi.com Amazon
USB-C power supply for monitor (optional, for full brightness)	5V/1.5A USB-C supply — skip if powering monitor from Pi USB port (Pi USB port gives 60% brightness which may be dim outdoors)	Amazon
— Premium Workstation (Pi 500+) —		
Raspberry Pi 500+ keyboard computer	Pi 500+ — Pi 5 chip, 16 GB RAM, 256 GB NVMe SSD, mechanical keyboard (Gateron Blue), RGB	raspberrypi.com Adafruit · Amazon
Raspberry Pi Monitor + cables + mouse	Same as Standard Workstation above	Same as above

501(c)(3) Grant Funding Note: MCEMV/MCEMA is a 501(c)(3) nonprofit. Operator workstation equipment may be eligible for funding through: FEMA BRIC (Building Resilient Infrastructure & Communities), ARPA-E Emergency Communications grants, ARRL Foundation grants, local Community Foundation equipment grants, or AUXCOMM/ARES regional funding. Document each workstation as "emergency communications operator terminal for the McHenry County Emergency Services Volunteers (K9ESV), a 501(c)(3) nonprofit supporting McHenry County Emergency Management Agency (MCEMA) RACES/ARES/Starcom communications."

Recommended Deployment by Position

POSITION	RECOMMENDED DEVICE	REASON
Net Control Operator (fixed EOC desk)	Pi 500 + Pi Monitor	Dedicated keyboard, good display, Chromium runs every FieldComms page smoothly
ICS Section Chief (Planning, Ops, Logistics)	Pi 500 + Pi Monitor or Windows/Mac laptop	Needs reliable keyboard for ICS form entry; Pi 500 is sufficient for all ICS platform pages
Served Agency Staff (EOC or shelter)	Any tablet, Chromebook, or Pi 500	Observer Mode and ICS forms work on any browser — no special hardware needed
Winlink / JS8Call Operator	Windows laptop (required)	Winlink Express and JS8Call are Windows applications — Pi 500 cannot run them natively

Mobile / Field Operator	Phone or tablet	FieldComms works on any smartphone browser on EMCOMM-NET — no dedicated hardware needed
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■ The Pi 500 cannot run Windows applications natively. Winlink Express and JS8Call must run on the Windows laptop connected to the IC-7300. The Pi 500 is ideal for all browser-based FieldComms tools — net control logging, ICS forms, tactical maps, resource tracking, and the full ICS platform.